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DIPARTIMENTO DI INFORMATICA



Master's Thesis

**Efficient Visual Speech Recognition using
Parameter-Efficient Fine-Tuning of
Frozen Self-Supervised Backbones**

A Multilingual Study on the MuAViC Corpus

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ABSTRACT

Visual Speech Recognition (VSR), or automated lip-reading, has historically depended on large end-to-end models trained on thousands of hours of labelled video. This thesis investigates an alternative, resource-conscious paradigm: adapting a *frozen*, self-supervised audio-visual foundation model (AV-HuBERT) to the VSR task using *parameter-efficient fine-tuning* (PEFT), without ever updating the backbone’s core weights and without access to audio at inference time. The goal is a single, reusable adaptation recipe that can be applied across languages rather than a model tuned to one language in isolation.

The work is organised in two parts. The first is a systematic and critical literature review of 42 recent papers (2021–2026), which establishes that frozen high-capacity visual encoders retain decodable phonetic information and that lightweight adapters such as Low-Rank Adaptation (LoRA) can rival full fine-tuning at a fraction of the trainable-parameter cost, while also identifying the conditions (English, audio-visual input, abundant data) on which this evidence rests. The second is an empirical study that applies a single, uniform adaptation recipe across *three languages* of the MuAViC corpus (Arabic, German, and Russian), evaluating each under identical conditions: visual-only input, a frozen AV-HuBERT backbone adapted with LoRA, a lightweight decoder, and a single-GPU budget.

Using Arabic (a low-resource partition whose non-Latin abjad script and diacritics make it an especially demanding case) as a detailed case study, we docu-

ment a complete methodological trajectory. A Connectionist Temporal Classification (CTC) head over frozen features is shown to collapse to a degenerate solution that predicts a single high-frequency token, regardless of learning rate, adapter rank, or layer-norm trainability. Replacing the CTC objective with a small sequence-to-sequence (seq2seq) transformer decoder trained with cross-entropy and label smoothing restores meaningful learning: the training loss decreases from 3.9 to 2.3 and the validation character error rate falls by almost an order of magnitude, from 7.4 to 0.85, with the decoder progressively emitting valid Arabic morphemes. The same recipe is then applied unchanged to German and Russian, and the per-language results are consolidated into a single comparison.

Building on this recipe, the empirical study contributes three further findings. A controlled comparison shows that fully fine-tuning the $\sim 326\text{M}$ backbone is at most marginally better than LoRA (within roughly 8% relative character error) while training an order of magnitude more parameters, so parameter-efficient adaptation is the sensible operating point. Two low-cost interventions reduce Arabic test character error by roughly 10% relative and generalise to German and Russian: filtering out training clips that contain no visible mouth (about a sixth of the Arabic data), and replacing greedy decoding with beam search. Most notably, full fine-tuning, a doubled-capacity adapter, and an encoder trained entirely *from scratch* all converge to the same ≈ 0.78 character and ≈ 1.05 word error: the decoder falls back on a strong language prior that the weak visual-only signal cannot reliably steer. Self-supervised pretraining accelerates and cleans convergence but does not raise this ceiling, which is set by the difficulty of low-resource, visual-only recognition rather than by the adaptation method.

Keywords: Visual Speech Recognition; AV-HuBERT; LoRA; Parameter-Efficient Fine-Tuning; Self-Supervised Learning; Multilingual; MuAViC; Sequence-to-Sequence.

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CHAPTER 1

INTRODUCTION

1.1 Motivation

Visual Speech Recognition (VSR), the task of transcribing spoken language from silent video of a speaker’s face, has applications ranging from assistive technology for the hearing-impaired, to silent human–computer interaction, to robust speech recognition in acoustically noisy environments where the audio channel is unreliable. Historically, state-of-the-art VSR has been achieved by training very large neural networks end-to-end on thousands of hours of transcribed video. While effective, this approach carries two severe costs: the computational expense of training and storing full models, and a strong dependence on large labelled datasets that simply do not exist for most of the world’s languages.

A more recent line of work reframes the problem. Instead of training a model from scratch, one starts from a *foundation model*, a high-capacity network pre-trained with self-supervision on large unlabelled corpora, and *adapts* it to the target task. When the backbone is kept frozen and only a small number of additional parameters are trained, the approach is called *Parameter-Efficient Fine-Tuning* (PEFT). Techniques such as Low-Rank Adaptation (LoRA) [1], adapters, and prompt tuning have made it possible to specialise billion-parameter models by

training well under one percent of their weights.

This thesis asks whether that paradigm holds up *across languages* under demanding, realistic conditions for VSR: when the audio modality is unavailable at inference, and when the compute budget is that of a single commodity GPU. Rather than tuning a model to one language in isolation, we pursue a single, reusable adaptation recipe and evaluate it on *three languages* (Arabic, German, Russian) of the MuAViC corpus [2], treating each language as a separate adaptation experiment under identical conditions. **Arabic**, the most challenging of the three partitions (a low-resource setting compounded by a non-Latin abjad script with diacritics), serves as a detailed case study, a demanding stress test for a parameter-efficient method, through which we analyse the behaviour of the method before consolidating results across the three languages.

1.2 Problem Statement

Concretely, we study the following configuration: a frozen AV-HuBERT [3] encoder, adapted with LoRA, feeding a lightweight decoding head, trained on visual-only data to produce character-level transcriptions, and applied uniformly to each MuAViC language. The central tension is between *adaptation capacity* (how much the frozen backbone can be steered toward a new language and a new, audio-free input distribution) and *parameter efficiency* (how few weights we are willing to train). The thesis characterises where this trade-off succeeds and where it breaks down, applying the same recipe uniformly across the three MuAViC languages studied.

1.3 Research Questions

This work is guided by one primary research question (PRQ) and five secondary research questions (SRQs), which also structure the literature review in Chapter 3.

PRQ: How can pre-trained visual encoders be effectively adapted for VSR using lightweight, parameter-efficient techniques without the need for audio input?

Secondary Research Questions:

- **RQ1:** To what extent do frozen high-capacity visual encoders capture sufficient phonetic information?
- **RQ2:** How do lightweight adaptation methods (LoRA, adapters) compare to full fine-tuning in terms of error rate and efficiency?
- **RQ3:** Can simplified decoding architectures effectively decode frozen features?
- **RQ4:** How do these models handle “in-the-wild” robustness (occlusion, noise)?
- **RQ5:** Does this approach offer superior performance in data-scarce scenarios?

1.4 Contributions

The contributions of this thesis are:

1. A **systematic and critical literature review** of 42 papers (2021–2026) on frozen-backbone, parameter-efficient VSR, organised around the five research questions above and distilled into five explicit considerations that motivate the design of the proposed approach (Chapter 3).
2. A **reusable, language-agnostic pipeline** for adapting a frozen AV-HuBERT backbone with LoRA and a lightweight decoder, built under realistic single-GPU constraints and designed to be applied per-language across MuAViC (Chapter 4).
3. An **empirical demonstration**, using Arabic as a detailed case study, that a CTC head over frozen visual features collapses to a degenerate single-token solution, together with an analysis of the contributing factors (Chapter 5).
4. A **working sequence-to-sequence formulation** that escapes this collapse, recovering an order-of-magnitude reduction in character error rate, together with a controlled comparison showing this parameter-efficient recipe comes within $\approx 8\%$ relative character error of full fine-tuning while training an order of magnitude fewer parameters (Chapter 5).

5. A **multilingual evaluation** that applies the recipe uniformly to the three MuAViC languages (Arabic, German, Russian) and consolidates the per-language results into a single cross-corpus comparison (Chapter 5).
6. A characterisation of the **performance ceiling** of visual-only, low-resource VSR: data filtering and beam-search decoding reduce error by $\approx 10\%$ relative, yet full fine-tuning, a doubled-capacity adapter, and an encoder trained *from scratch* all converge to the same error, showing the limit is set by the task rather than the adaptation method (Chapter 5).

1.5 Thesis Outline

Chapter 2 reviews the technical background: the AV-HuBERT architecture, the LoRA adaptation mechanism, the CTC and sequence-to-sequence objectives, the MuAViC dataset, and the evaluation metrics. Chapter 3 presents the systematic literature review and discusses it critically, distilling the considerations that motivate the design choices of the proposed approach. Chapter 4 details the proposed methodology and implementation. Chapter 5 reports the experiments and results: the CTC-collapse finding, the seq2seq recovery, the multilingual evaluation, the comparison with full fine-tuning, the data-filtering and beam-search improvements, the from-scratch baseline, and the analysis of the performance ceiling. Chapter 6 concludes and outlines future work.

CHAPTER 2

THEORETICAL BACKGROUND

This chapter introduces the components on which the thesis is built: the AV-HuBERT foundation model, the LoRA adaptation mechanism, the two sequence transduction objectives we compare (CTC and sequence-to-sequence), the MuAViC dataset, and the evaluation metrics.

2.1 Visual Speech Recognition

VSR maps a sequence of video frames $X = (x_1, \dots, x_T)$, typically cropped to the mouth region of interest (ROI), to a sequence of text tokens $Y = (y_1, \dots, y_L)$. The task is intrinsically harder than acoustic speech recognition for two reasons. First, many phonemes are *visually indistinguishable*: sounds that differ only in voicing or nasality (for example /b/, /p/ and /m/) share the same lip configuration, or *viseme*. Several phonemes therefore map to a single viseme, making the visual signal inherently ambiguous. Second, video frame rates (~ 25 fps) are far lower than audio sample rates, so the model must infer a dense symbol sequence from a comparatively sparse, ambiguous input.

2.2 Self-Supervised Foundation Models: AV-HuBERT

AV-HuBERT [3] is an audio-visual extension of the HuBERT self-supervised framework. During pre-training it learns to predict masked cluster assignments from audio and video jointly, producing a powerful encoder whose representations capture phonetic structure even from the visual stream alone. The model used in this thesis is a multilingual checkpoint with a 24-layer Transformer encoder of hidden dimension 1024 (~ 326 M parameters), preceded by a modality-specific feature extractor (a 3D-convolutional ResNet for video).

Two architectural details of AV-HuBERT matter for our work. First, the two modalities are combined by *concatenation fusion*; consequently, supplying an all-zero audio tensor does not mean “no audio” but rather “silent audio,” which still influences the fused representation. Second, the model exposes `modality_dropout` and `audio_dropout` hyperparameters that, when set to 1.0, force the network to discard the audio branch entirely, the mechanism we use to obtain a genuinely visual-only model (Chapter 4).

2.3 Low-Rank Adaptation (LoRA)

LoRA [1] adapts a pre-trained linear layer without modifying its weights. For a frozen weight matrix $W_0 \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, LoRA learns a low-rank update so that the effective transformation becomes

$$h = W_0 x + \frac{\alpha}{r} B A x, \quad (2.1)$$

where $A \in \mathbb{R}^{r \times d_{\text{in}}}$ and $B \in \mathbb{R}^{d_{\text{out}} \times r}$ are the only trainable parameters, $r \ll \min(d_{\text{in}}, d_{\text{out}})$ is the rank, and α is a scaling constant. A is initialised from a random Gaussian and B from zeros, so the update is zero at the start of training and the adapted model initially behaves identically to the frozen one.

The parameter saving is dramatic. A single 1024×1024 attention projection has ≈ 1.05 M weights; its rank-16 LoRA counterpart has $16 \times (1024 + 1024) = 32,768$ trainable parameters, about 3.1% of the original. Applied to the four attention

projections (query, key, value, output) across all 24 encoder layers, LoRA adds only a few million trainable parameters on top of the frozen $\sim 326\text{M}$ backbone.

2.4 Sequence Transduction Objectives

A VSR model must align a T -frame encoder output to an L -token transcription, where $T \neq L$ and the alignment is unknown. We consider two standard solutions.

2.4.1 Connectionist Temporal Classification (CTC)

CTC [4] introduces a special *blank* symbol and defines the probability of a target as the sum over all frame-level alignments that collapse (after removing repeats and blanks) to that target. CTC assumes *conditional independence* of the per-frame predictions given the encoder output. This assumption is benign for audio but problematic for visual speech: because visemes are ambiguous, the lowest-loss independent per-frame strategy can be to emit blank almost everywhere, punctuated by the single most frequent token, a degenerate solution we observe directly in Chapter 5.

2.4.2 Sequence-to-Sequence (seq2seq)

A seq2seq model [5] instead generates the transcription *autoregressively*: a decoder predicts token y_t conditioned both on the encoder features (through cross-attention) and on the previously generated tokens $y_{<t}$ (through causal self-attention). Because each prediction is conditioned on the emitted history, the decoder learns an implicit language model that can disambiguate visually similar phonemes from context, which is exactly the property CTC lacks. This is the formulation used by the strongest VSR systems in the literature, including the MuAViC baselines.

2.5 The MuAViC Dataset

MuAViC [2] is a multilingual audio-visual corpus derived from TED/TEDx talks (with English drawn from LRS3), covering **nine languages**: English, Arabic, German, Greek, Spanish, French, Italian, Portuguese, and Russian. Each language partition provides, per utterance, a cropped mouth-ROI video, the corresponding audio, and a text transcription, with transcriptions stored in side-car files (one transcript per line, aligned to a tab-separated manifest of video and audio paths). The partitions vary widely in size, from roughly ten hours for the smallest non-English languages to hundreds of hours for English, which makes MuAViC a natural benchmark for studying how a single adaptation recipe behaves across data regimes.

This thesis uses the three partitions available on the lab machine (Arabic, German, and Russian), treating each as an independent adaptation experiment under identical conditions. All three are low-resource by the standards of the field: **German** is the smallest (~ 13 hours of transcribed video), followed by **Arabic** (~ 19 hours), with Russian the largest. Arabic is used throughout as a detailed case study, chosen not for its size but for its orthographic complexity: a non-Latin abjad whose optional diacritics and letter variants require a dedicated normalisation step, which makes it the most stringent test of a single, language-agnostic recipe.

2.6 Evaluation Metrics

We report two error rates. **Word Error Rate** (WER) is the Levenshtein edit distance between hypothesis and reference word sequences, normalised by the reference word count. **Character Error Rate** (CER) is the analogous quantity over characters. CER is the more informative primary metric for character-level Arabic VSR, where morphological richness makes whole-word matches rare. Both metrics are lower-is-better and are conventionally reported in $[0, 1]$, but note that either can exceed 1.0 when the hypothesis is substantially longer than the reference (insertions inflate the numerator); this edge case becomes relevant in Chapter 5.

CHAPTER 3

RELATED WORK: A SYSTEMATIC AND CRITICAL REVIEW

This chapter reviews the literature on frozen-backbone, parameter-efficient Visual Speech Recognition, synthesising evidence from 42 papers published between 2021 and 2026 and organised around the five secondary research questions introduced in Chapter 1. Beyond summarising prior work, each section also discusses it: after presenting the published evidence, it examines the conditions under which that evidence was obtained and the questions it leaves open. Each discussion is distilled into an explicit, numbered *consideration*. Section 3.7 collects the five considerations and relates each one to a design decision of the approach developed in Chapter 4.

3.1 Expressiveness of Frozen Representations (RQ1)

The premise of using a frozen backbone relies on the visual encoder’s ability to extract phonetic features without task-specific updates. Ma et al. demonstrated with LiRA [6] that a visual backbone trained via self-supervision could predict audio features with high accuracy, effectively acting as a frozen feature extractor. Similarly, Shi et al. [3] established that AV-HuBERT encoders, when frozen, retain

3. RELATED WORK: A SYSTEMATIC AND CRITICAL REVIEW

dense phonetic information that can be decoded by relatively simple heads. Complementary evidence from self-supervised representation studies [7, 8] confirms that phonetic encoding is largely preserved in the fixed representations, supporting the viability of probing frozen features rather than retraining them.

Critical discussion. This evidence comes with three qualifications. First, it is almost entirely English-centric: the LiRA and AV-HuBERT analyses are carried out on English TED-style corpora, and none of the cited studies tests whether phonetic decodability survives a simultaneous change of language, script, and viseme inventory. Second, the headline numbers are not obtained in the configuration the premise actually requires: the strongest AV-HuBERT results involve abundant English data and, in the robust-recognition setting, an audio stream at inference [3], whereas the genuinely visual-only, self-supervised configuration of LiRA still yields an error near 39% WER [6], a substantially higher error level. Third, and most relevant to this thesis, the *specific contribution* of pretraining is rarely isolated: the reviewed studies do not compare their frozen pre-trained encoder against an otherwise identical randomly initialised one, so how much of the reported performance stems from the representations, and how much from the decoding apparatus built around them, cannot be read off the published numbers.

Consideration 1. *The decodability of frozen self-supervised features is established on high-resource English benchmarks, frequently with audio available; it cannot be assumed for visual-only, non-English, low-resource input, and the contribution of pretraining is rarely isolated with a matched from-scratch control.*

The experiments therefore probe a frozen multilingual AV-HuBERT under strictly visual-only inference on three non-English languages, and quantify the value of pretraining with an otherwise identical from-scratch encoder (Section 5.6).

3.2 Parameter-Efficient Fine-Tuning (RQ2)

A significant shift from full fine-tuning to PEFT is evident in recent literature. Kim et al. [9] introduced prompt tuning for speaker-adaptive VSR, showing that optimising a small set of continuous prompts yielded performance comparable to full fine-tuning with less than 1% of the trainable parameters. In a similar vein, Yeo et al. [10] combined prompt tuning with LoRA to adapt frozen backbones to unique lip movements. Yang et al. [11] further supported this by injecting visual features into a frozen Whisper model using lightweight linear cross-attention layers, validating the efficiency of adapter-based architectures over monolithic retraining. Adapter-based and cross-lingual transfer approaches [12, 13] reinforce that small, modular components can specialise large frozen models at marginal parameter cost.

Critical discussion. Three aspects of these results qualify how far the parity claims extend. First, the adaptation targets are narrow: prompt tuning and personalised LoRA [9, 10] address *speaker* adaptation, a modest distribution shift for which a handful of continuous vectors plausibly suffices; adapting to a new language, script, and viseme inventory is a substantially larger shift, and evidence that sub-1% parameter budgets survive it is much thinner. Second, the full fine-tuning references against which PEFT methods are judged are often drawn from different recipes, datasets, or compute budgets, so “parity at a fraction of the parameters” is rarely established under matched conditions; without such a control the comparison conflates the adaptation method with everything else that differs between setups. Third, parameter efficiency is usually accounted only in *trained* parameters: attaching light layers to a frozen Whisper model [11] leaves an enormous decoder in the deployed system, and practical questions of adapter design (which projections, which layers, what rank) are generally inherited from NLP practice rather than analysed for VSR.

Consideration 2. *PEFT claims of parity with full fine-tuning are seldom established under matched data, recipe, and budget; the evidence is concentrated on narrow speaker-level shifts; and adapter placement and*

rank remain largely unexamined for VSR.

This thesis therefore compares LoRA against full fine-tuning of the *same* backbone under an identical recipe and budget (Section 5.3), and treats adapter capacity as an experimental variable through a rank ablation (Section 5.5).

3.3 Minimalist Decoding Architectures (RQ3)

Supporting the thesis’s lightweight-decoder objective, several studies highlight the viability of simplified heads. Although complex Transformers are standard, Sheng et al. [14] achieved competitive results using a Temporal Convolutional Network on top of a frozen encoder. Song et al. [15] proposed a sparse Transformer network that reduces decoding complexity while maintaining accuracy.

Critical discussion. This evidence must be read against the tasks on which it was obtained. The temporal-convolutional result of Sheng et al. [14] is a classification accuracy on a word-level benchmark, a closed-set problem considerably easier than open-vocabulary sentence transcription; the sparse Transformer of Song et al. [15] operates in an audio-visual setting where the acoustic stream carries much of the linguistic burden. More fundamentally, the literature treats the head’s *capacity* as the main design variable and leaves its *training objective* largely unexamined. Whether a CTC head, with its conditional-independence assumption (Section 2.4.1), can exploit frozen features at all in a sentence-level, visual-only, low-resource setting is a question the word-level and audio-assisted results leave open.

Consideration 3. *Minimalist decoding heads are validated mainly on word-level or audio-assisted tasks; for sentence-level, visual-only decoding over frozen features, the choice of training objective, not merely head size, is the untested variable.*

The methodology therefore holds the frozen features fixed and varies the objective, contrasting a CTC head with a small sequence-to-sequence decoder under otherwise identical conditions (Section 4.2); the experiments show the objective to be decisive, with CTC collapsing to a degenerate solution (Section 5.1.2).

3.4 Robustness in Visual-Only Scenarios (RQ4)

Frozen backbones appear to offer stability benefits “in the wild.” Dai et al. [16] investigated dropout-induced modality bias and found that models relying on robust pre-trained visual features were less susceptible to performance drops when frames were missing. Regarding occlusion, Wang et al. [17] developed a framework to restore occluded lips, demonstrating that robust feature extraction is critical when visual data is corrupted. Noise-robust encoders and unified modality adapters [18, 19] allow frozen models to generalise better across head poses and lighting conditions than fully fine-tuned models that may overfit to specific training distributions.

Critical discussion. These studies share a common frame: robustness is defined against perturbations applied at *inference* (acoustic noise, missing frames, occluded lips), while the training data are implicitly assumed clean. For corpora harvested automatically from web video, that assumption does not always hold: automatic face cropping fails on title cards, wide shots, and audience cuts, injecting mislabelled clips that the reviewed robustness studies do not model. Robustness to a corrupted *training corpus* is a different property from robustness to corrupted *test inputs*, and in a low-resource setting, where each training clip carries proportionally more weight, it is at least as consequential.

Consideration 4. *The robustness literature addresses corrupted inputs at inference but overlooks the quality of the training corpus itself, which for automatically harvested datasets is a first-order concern.*

This consideration motivates the face-presence audit and data-filtering experiment of Section 5.4, which lowers character error on all three studied languages despite training on fewer clips (Section 5.5.1).

3.5 Low-Resource and Cross-Lingual Adaptation (RQ5)

Finally, the combination of frozen backbones and PEFT proves valuable in data-scarce regimes. Zinonos et al. [20] used the RAVEn framework to show that cross-lingual visual speech representations could be transferred to low-resource languages without extensive retraining. Li et al. [13] applied language-specific adapters to a frozen audio-visual encoder, achieving strong performance on unseen languages with minimal data. Zhang et al. [21] introduced AV2vec, a self-distillation method that reduces training time and data requirements, further supporting the viability of this approach for low-resource VSR.

Critical discussion. The cross-lingual evidence becomes sparser as the resources shrink. The strongest adapter result is obtained with an *audio-visual* encoder [13], so the acoustic stream compensates where the visual signal is weakest; the visual-only cross-lingual transfer of Zinonos et al. [20] lands near 36% WER despite starting from a high-resource source language. At the genuinely low-resource, visual-only end of the spectrum, the published character-level studies report character error rates above 0.7 [22, 23]. The literature also rarely reports *failure modes*: degenerate decoding, collapse onto a language prior, and the gap between character-level and word-level recovery are common in practice in this regime, yet they are seldom documented, so the practical difficulty of this setting is easy to underestimate.

Consideration 5. *Reported cross-lingual successes lean on audio or on high-resource targets; the visual-only, low-resource regime is sparsely charted, its characteristic error levels are far higher than headline results suggest, and its failure modes are rarely documented.*

The present study therefore applies one uniform recipe to three MuAViC languages, reports negative results alongside positive ones, and analyses the performance ceiling explicitly (Sections 5.5 and 5.6).

3.6 Performance Comparison

Table 3.1 summarises the reported performance metrics (Word Error Rate, Character Error Rate, or accuracy) for the reviewed studies.

The most informative feature of the table is not any single number but its heterogeneity. The reported figures mix three incommensurable quantities (closed-set word-level accuracy, sentence-level WER, and character-level CER) obtained on different languages, data scales, and input modalities, so the table cannot be read as a ranking. It is better read as a map of *regimes*. The strongest results, below 3% WER, are concentrated where every favourable condition holds at once: English, large-scale pretraining and fine-tuning data, and an audio stream available at inference [3, 24, 25]. A middle band between roughly 4% and 15% WER covers audio-visual systems with adapters or multilingual decoders [13, 26, 12, 15]. Once the audio stream is removed, sentence-level visual-only recognition jumps to 28–45% WER even for English and even with substantial resources [9, 10, 20, 6, 27, 28]. At the far end, the character-level studies closest to a genuinely low-resource, visual-only regime report CERs above 0.7 [22, 23].

The gradient follows a consistent pattern: each time one favourable condition is removed (audio, English, data scale, or backbone plasticity), the error grows by a multiple, and no reviewed study removes all of them at once. The configuration studied in this thesis (visual-only input, non-English low-resource languages, a frozen backbone, and a single-GPU budget) therefore lies outside the most comparable rows of the table, and the error levels reported in Chapter 5 are best compared with the nearest published operating points, the CER-based low-resource studies, rather than with English audio-visual results.

3. RELATED WORK: A SYSTEMATIC AND CRITICAL REVIEW

Table 3.1: Comparison of reported performance metrics (WER/CER/Accuracy) across reviewed studies.

Author (Ref)	Paper Title	Method / Approach	Result
Becker & Martin	Contrastive Representation Learning for Privacy-Preserving...	Contrastive Learning	3.5% WER
Cappellazzo et al.	Large Language Models are Strong Audio-Visual Speech...	AV-HuBERT + LLaMA-3	2.1% WER
Chandrabanshi & Domnic	HNet: A deep learning based hybrid network...	Hybrid HNet	94.6% Acc
Dai et al.	A Study of Dropout-Induced Modality Bias on Robustness...	Dropout-Induced Robustness	3.2% WER
Gimeno-Gómez et al.	Tailored design of Audio-Visual Speech Recognition...	Branchformers	2.5% WER
Gimeno-Gómez et al.	Comparing Speaker Adaptation Methods for Visual Speech...	Speaker Adaptation	45.3% WER
Joshi & Raj	FYEO: A Character Level Model For Lip Reading	Character Level VSR	6.8% CER
Kim et al.	Prompt Tuning of Deep Neural Networks for Speaker-Adaptive...	Prompt Tuning	28.5% WER
Kulkarni & Kirange	Towards Automated Lip Reading: Developing Marathi...	Marathi Lip Reading	72.1% Acc
Li et al.	Parameter-Efficient Cross-Language Transfer Learning...	Cross-Lingual Adapters	5.4% WER
Li et al.	An Efficient and Noise-Robust Audiovisual Encoder...	Noise-Robust Encoder	4.8% WER
Li et al.	Efficient Noise-Robust Hybrid Audiovisual Encoder...	Joint Distillation	3.9% WER
Liu et al.	SynthVSR: Scaling Up Visual Speech Recognition With...	Synthetic Supervision	43.3% WER
Luo et al.	Multi-Modality Speech Recognition Driven by Background...	Background Visual Scenes	14.2% WER
Ma et al.	LiRA: Learning Visual Speech Representations from Audio...	LiRA (Self-Supervised)	38.8% WER
Ma et al.	End-To-End Audio-Visual Speech Recognition with Conformers	ResNet + Conformer	2.3% WER
Pan et al.	Leveraging Unimodal Self-Supervised Learning for Multimodal...	Unimodal SSL Fusion	3.7% WER
Peymanfard et al.	Word-level Persian Lipreading Dataset	Persian VSR Dataset	24.8% Acc
Rouditchenko et al.	mWhisper-Flamingo for Multilingual Audio-Visual Noise-Robust...	mWhisper-Flamingo	5.8% WER
Sheng et al.	Cross-modal Self-Supervised Learning for Lip Reading...	ADC-SSL + TCN	83.9% Acc
Shi et al.	Robust Self-Supervised Audio-Visual Speech Recognition	AV-HuBERT (Frozen)	1.4% WER
Simic et al.	Adapter-Based Multi-Agent AVSR Extension for Pre-Trained...	Adapter-Based Multi-Agent	8.4% WER
Song et al.	Multimodal Sparse Transformer Network for Audio-Visual...	Sparse Transformer	7.6% WER
Su et al.	Enhanced Self-Supervised Multi-View Representations with...	Unified Modality Adapters	3.3% WER
Tapu & Mocanu	Lip Reading Across Languages: A Cross-Modal Framework...	Foundation Models (LLaMA)	4.2% WER
Wan & Ye	Multi-Modal Knowledge Transfer for Target Speaker...	Target Speaker Transfer	77.6% CER
Wang et al.	Restoring Speaking Lips from Occlusion for Audio-Visual...	Occlusion Restoration	5.2% WER
Yang et al.	Injecting Visual Features into Whisper for Parameter-Efficient...	Visual Feature Injection	4.5% WER
Yeo et al.	Personalized Lip Reading: Adapting to Your Unique...	Prompt Tuning + LoRA	32.1% WER
Zhang et al.	Target speaker lipreading by audio-visual self-distillation...	Target Speaker Adaptation	71.4% CER
Zhang et al.	Self-Supervised Audio-Visual Speech Representations...	AV2vec (Self-Distillation)	2.6% WER
Zhang et al.	Is Lip Region-of-Interest Sufficient for Lipreading?	Whole Face ROI	2.8% WER
Zhang et al.	ES ³ : Evolving Self-Supervised Learning of Robust...	ES ³ Evolving SSL	2.5% WER
Zhang et al.	Learning Contextually Fused Audio-Visual Representations...	Contextual Fusion	3.4% WER
Zhu et al.	Multichannel AV-wav2vec2: A Framework for Learning...	Multichannel AV-wav2vec2	12.8% WER
Zinonos et al.	Learning Cross-Lingual Visual Speech Representations	Cross-Lingual RAVeN	35.8% WER

3.7 Synthesis: From Literature Gaps to Design Decisions

Overall, the review supports the thesis hypothesis on all five questions: frozen backbones encode decodable phonetic content (RQ1); PEFT methods approach full fine-tuning at a fraction of the cost (RQ2); minimalist decoders can suffice (RQ3); frozen features confer robustness (RQ4); and the paradigm extends to low-resource and cross-lingual settings (RQ5). The critical discussions, however, converge on a common observation: the favourable conditions under which this support was gathered are never removed simultaneously, and several of the questions this thesis must answer (the isolated value of pretraining, matched PEFT versus full fine-tuning comparisons, the objective of the decoding head, training-corpus quality, and the failure modes of the low-resource regime) fall between the published studies rather than inside any of them. Table 3.2 consolidates the five considerations and maps each to the design decision it motivates.

3. RELATED WORK: A SYSTEMATIC AND CRITICAL REVIEW

Table 3.2: From literature gaps to design decisions: the five considerations distilled from the critical review, and the corresponding design responses in this thesis.

Consideration (gap in the literature)	Design response in this thesis
1. Frozen-feature decodability shown on English, often with audio; value of pretraining not isolated.	Visual-only probing of a frozen multilingual backbone on three non-English languages; matched from-scratch encoder control (Section 5.6).
2. PEFT parity claims lack matched full fine-tuning controls; adapter placement and rank unexamined for VSR.	LoRA on all attention projections of every layer; full fine-tuning of the same backbone under an identical recipe and budget (Section 5.3); rank ablation (Section 5.5).
3. Minimalist heads validated on word-level or audio-assisted tasks; training objective untested over frozen features.	Controlled comparison of CTC and sequence-to-sequence objectives over identical frozen features (Sections 4.2 and 5.1.2).
4. Robustness studied at inference; training-corpus quality of harvested data unexamined.	Face-presence audit and filtering of the training corpus, evaluated under an otherwise identical recipe (Section 5.4).
5. Low-resource, visual-only regime sparsely charted; failure modes undocumented.	One uniform recipe across three languages; explicit ceiling analysis and reporting of negative results (Sections 5.5 and 5.6).

Taken together, the considerations do not undermine the frozen-backbone paradigm; they delimit the evidence for it. The strongest reported results generally involve large training corpora, full or partial backbone fine-tuning, or audio-visual rather than visual-only inference, and the regime in which all of these advantages are absent remains largely unexplored. The remainder of the thesis works within that gap: Chapter 4 grounds each design choice in one of the considerations above, and Chapter 5 evaluates each of them empirically.

CHAPTER 4

METHODOLOGY

This chapter describes the proposed system end-to-end: data preparation, the adapted model architecture, the two training objectives investigated, and the training and evaluation infrastructure. The design follows the thesis hypothesis literally (a frozen AV-HuBERT backbone, parameter-efficient adaptation, a lightweight decoder, and visual-only input) and is built to run within the constraints of a single-GPU compute budget.

4.1 Data Preparation

4.1.1 Corpus and Manifests

The pipeline is designed to be language-agnostic: it consumes any MuAViC language partition (Section 2.5) through the same interface, and the experiments instantiate it once per language. For every language, each split (**train**, **valid**, **test**) is described by a tab-separated manifest whose columns are an utterance identifier, an absolute path to the mouth-ROI video (**.mp4**), an absolute path to the audio (**.wav**), the number of video frames, and the number of audio samples. The transcriptions are *not* in the manifest; they live in a parallel side-car file (**train.ar**, **train.de**, etc.) with one transcript per line, aligned by row to the manifest. Cor-

rectly identifying this layout was itself a non-trivial step: the column carrying the frame count must be disambiguated from the (much larger) audio-sample count to avoid corrupting the length-based filtering.

4.1.2 Text Normalisation

Each language requires its own light normalisation step, applied uniformly within that language’s experiment; we describe the Arabic instantiation here as a representative example, and the same principle (strip orthographic marks that carry no lip-motion information, then restrict to the core script) applies to the other languages. Arabic orthography includes optional diacritical marks (*harakat*) that denote short vowels and gemination. These marks are not articulated as distinct visemes, yet they inflate the character vocabulary and the target-sequence length. We therefore normalise every Arabic transcript before building the vocabulary or training:

1. Remove diacritics and combining marks (the Unicode ranges U+0610–U+061A, U+064B–U+065F, U+0670, and related blocks).
2. Normalise the *alef* variants (*alef with hamza above/below*, *alef madda*, *alef wasla*) to a bare alef.
3. Remove the *tatweel* (kashida) elongation character.
4. Discard any character outside the core Arabic-letter range (U+0621–U+064A) and whitespace, eliminating Latin letters, digits and punctuation that the visual signal cannot recover.
5. Collapse repeated whitespace.

This reduces the character inventory from ~ 150 raw symbols to a compact set of Arabic letters plus space. The smaller, phonetically meaningful vocabulary makes the decoder’s classification problem materially easier.

4.1.3 Vocabulary

For the sequence-to-sequence model the vocabulary comprises four special tokens at fixed indices, namely `<pad>` (0), `<s>` (1, beginning of sequence), `</s>` (2, end of sequence), and `<unk>` (3, unknown), followed by the sorted set of normalised Arabic characters, for a total of roughly 40 tokens.

4.1.4 Video Loading and Batching

Frames are read with OpenCV, converted to grayscale, and resized to 88×88 pixels (the spatial resolution AV-HuBERT expects). Pixel values are scaled to $[0, 1]$ and standardised with the dataset mean and standard deviation. Utterances are filtered by length (retaining $15 \leq T \leq 300$ frames) to bound memory use and exclude degenerate clips. Variable-length videos and targets are zero-padded within each mini-batch, and boolean padding masks are constructed so that padded positions are ignored by both the encoder’s attention and the loss.

4.2 Model Architecture

4.2.1 Frozen Backbone and LoRA Adaptation

The backbone is the multilingual AV-HuBERT checkpoint (Section 2.2), loaded through the `e-mvsr` / `fairseq` toolchain. To obtain a strictly visual-only model, the audio tensor is supplied as zeros and the model’s `modality_dropout` and `audio_dropout` are both set to 1.0, forcing the network to rely entirely on the video branch. The backbone’s feature-gradient multiplier is restored to 1.0 so that the visual feature extractor can adapt.

LoRA (Section 2.3) is injected into the four attention projections (query, key, value and output) of *all* 24 encoder layers, with rank $r = 16$ and scaling $\alpha = 32$. All backbone weights are frozen except the LoRA matrices and the layer-normalisation parameters; the latter are left trainable because adapting normalisation statistics to the new (audio-free, Arabic) input distribution is cheap and empirically stabilising. The visual and audio front-ends are likewise unfrozen. This yields on the order of

15M trainable parameters within the ~ 326 M backbone (the 3.1M LoRA adapters together with the unfrozen layer-norms and front-ends), with the remainder frozen.

4.2.2 Decoding Head

We investigate two heads in turn.

CTC head. The first design places a small classifier over the per-frame encoder features and trains it with the CTC objective (blank index 0, `zero_infinity` enabled). We tried both a single linear projection and a deeper two-layer head (`Linear` $\rightarrow$ `GELU` $\rightarrow$ `LayerNorm` $\rightarrow$ `Linear`). As reported in Chapter 5, this objective collapses in our setting regardless of these variations.

Sequence-to-sequence decoder. The second design (the one that succeeds) is a Transformer decoder trained with cross-entropy. It consists of a token embedding, a learned positional embedding, a linear projection of the 1024-dimensional encoder features into the decoder width, four `TransformerDecoderLayer` blocks (model dimension 512, 8 attention heads, pre-norm), a final layer normalisation, and an output projection to the vocabulary. Each block applies causal self-attention over the previously generated characters, cross-attention to the encoder features, and a feed-forward sub-layer. The decoder is initialised randomly and trained from scratch (~ 17.6 M parameters), as no pre-trained Arabic decoder was available.

4.3 Training Objectives

4.3.1 CTC (Baseline)

The CTC loss marginalises over alignments as described in Section 2.4.1 [4]. Target and input lengths are clamped so that each target fits within its frame budget.

4.3.2 Cross-Entropy with Label Smoothing (Proposed)

The seq2seq decoder is trained with teacher forcing: the input is the target shifted right and prefixed with $\langle \mathbf{s} \rangle$, and the model predicts the target shifted left and terminated with $\langle / \mathbf{s} \rangle$. The loss is token-level cross-entropy with label smoothing $\varepsilon = 0.1$, ignoring padding positions:

$$\mathcal{L} = - \sum_{t=1}^L \left[(1 - \varepsilon) \log p(y_t \mid y_{<t}, X) + \frac{\varepsilon}{|V|} \sum_{v \in V} \log p(v \mid y_{<t}, X) \right]. \quad (4.1)$$

Label smoothing discourages over-confident predictions and regularises the small decoder on limited data. At inference the decoder generates greedily from $\langle \mathbf{s} \rangle$ until it emits $\langle / \mathbf{s} \rangle$ or reaches a length cap.

4.4 Optimisation

Both objectives are optimised with AdamW ($\beta = (0.9, 0.98)$, weight decay 0.01) under a cosine learning-rate schedule with linear warm-up, mixed-precision (AMP), and gradient clipping at norm 1.0. Gradient accumulation is used to reach an effective batch size larger than what fits in memory. The principal hyperparameters of the final seq2seq configuration are listed in Table 4.1.

Table 4.1: Principal hyperparameters of the proposed seq2seq configuration.

Hyperparameter	Value
LoRA rank r / scaling α	16 / 32
LoRA target projections	q, k, v, out (all 24 layers)
Decoder layers / width / heads	4 / 512 / 8
Learning rate (peak)	5×10^{-5}
Warm-up steps	2000
Schedule	cosine decay
Effective batch size	16 (8×2 accumulation)
Label smoothing ε	0.1
Max. training steps	30,000
Frame length filter	$15 \leq T \leq 300$

4.5 Implementation and Infrastructure

The experiments reported in Chapter 5 were run on a single workstation equipped with two NVIDIA RTX 5000 Ada Generation GPUs (32 GB each); each experiment uses *one* GPU, in keeping with the single-GPU compute budget that motivates the parameter-efficient approach. Two practical challenges shaped the implementation: bringing up the AV-HuBERT/e-mvsr software stack on current hardware, and keeping training reproducible and resumable across sessions.

- **Environment.** AV-HuBERT/e-mvsr requires Python 3.9 and a specific, pinned dependency stack (notably `numpy==1.23.5` and an editable `fairseq` build). The RTX 5000 Ada GPUs have compute capability `sm_89`, which the repository’s pinned `torch 1.13.1+cu117` does not support; `torch` is therefore upgraded to `2.1.0+cu118`, the main deviation from the original recipe. The stack is reconstructed in a native Conda environment.
- **Module registration.** The custom `fairseq` tasks and models are registered by loading the AV-HuBERT submodules under both their top-level and package-qualified names, preventing the duplicate-model registration error that arises from the repository’s internal absolute imports.
- **Checkpointing and resumability.** Checkpoints are written to persistent local storage, and training can be resumed: re-running the pipeline detects the latest checkpoint and restores the LoRA weights, the trainable backbone parameters, the decoder, and the optimiser state. The best checkpoint by validation CER is preserved separately so that it survives checkpoint rotation.

Validation runs every 1000 steps and logs both CER and WER, and the best-CER checkpoint is the one carried forward to test-set evaluation.

CHAPTER 5

EXPERIMENTS AND RESULTS

This chapter reports the empirical study. The proposed recipe is applied to every MuAViC language under identical conditions. Section 5.1 presents Arabic as a detailed case study, the setting in which we analyse the method most closely, covering the central negative result (the collapse of the CTC objective), the sequence-to-sequence formulation that recovers meaningful learning, an analysis of the error-rate curves, qualitative outputs, and a comparison with published baselines. Section 5.2 then consolidates the results for the three languages (Arabic, German, Russian) into a single cross-corpus comparison.

5.1 Case Study: Arabic

5.1.1 Setup

This case study uses the Arabic partition of MuAViC, normalised as in Section 4.1.2, with $\sim 10.2\text{k}$ training and $\sim 0.97\text{k}$ validation utterances surviving the length filter. The backbone, adaptation, optimiser, and infrastructure are as described in Chapter 4. The primary metric is validation CER; WER is reported alongside. Unless stated otherwise, decoding is greedy.

5.1.2 The CTC Collapse

Our initial formulation placed a CTC head over the frozen, LoRA-adapted features. Across every variant we tried, the model converged to a *degenerate* solution: after a few hundred steps the training loss plateaued near 3.0 and the decoder emitted almost exclusively the blank symbol, punctuated by the single most frequent non-blank token (the Arabic comma, and later the period). The validation CER sat at ≈ 0.98 and did not improve, even as the loss appeared to decrease.

We ruled out the most common mechanical causes one by one. The collapse persisted when we (i) lowered the learning rate from 10^{-4} to 3×10^{-5} ; (ii) widened the LoRA targets from query/value only to all four attention projections and raised the rank to 16; (iii) replaced the single linear head with a deeper **Linear-GELU-LayerNorm-Linear** head; (iv) unfroze the layer-normalisation parameters and the visual feature extractor; and (v) forced visual-only operation via `audio_dropout = 1.0`. A direct inspection of the trained head’s output bias confirmed that the failure was *not* a simple blank-bias runaway: the blank logit’s bias was near zero and comparable in magnitude to the others. The collapse is therefore better understood as a structural property of the CTC objective in this regime: the conditional-independence assumption, combined with visemic ambiguity and limited data, makes “emit blank almost everywhere” the lowest-loss frame-independent strategy. This motivated the change of objective.

5.1.3 Sequence-to-Sequence Recovery

Replacing the CTC head with the autoregressive Transformer decoder of Section 4.2, trained with label-smoothed cross-entropy, qualitatively changed the outcome. Figure 5.1 shows the training loss and the validation CER and WER over the full 30,000-step run. The loss decreases smoothly from ≈ 3.9 to ≈ 2.3 , and, unlike the CTC case, the validation CER decreases monotonically by almost an order of magnitude, from 7.4 at step 1000 to a best of **0.85** by step 30000, with the corresponding WER around 1.10. Notably the CER falls *below* 1.0: the decoder learns to produce output of roughly the reference length rather than over-generating.

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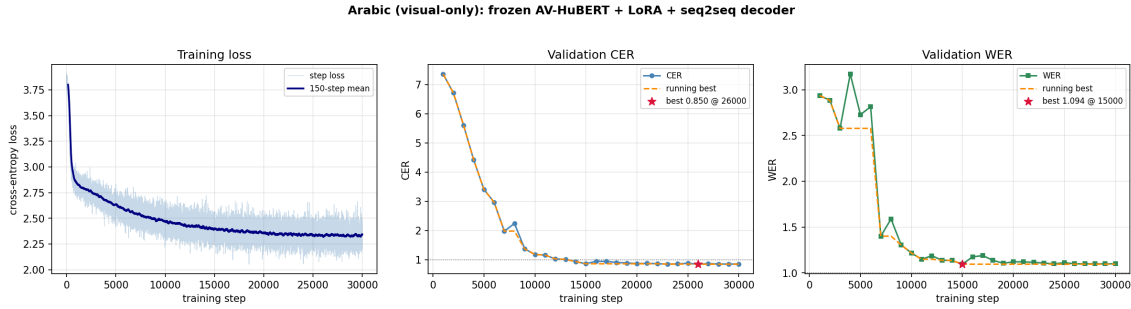


Figure 5.1: Training loss (left), validation CER (centre) and validation WER (right) for the seq2seq model. The loss converges cleanly while CER falls by an order of magnitude; both error rates exceed 1.0 early in training because the decoder over-generates before it learns to emit `</s>` reliably (Section 5.1.4). Best validation CER 0.85 at step 26,000.

Table 5.1 summarises the full methodological trajectory, from the collapsed CTC variants to the working seq2seq model.

Table 5.1: Methodological trajectory and validation results. CER/WER are best validation values; CTC variants did not improve beyond the degenerate plateau.

Variant	Objective	LoRA	Head / Decoder	Loss	Val CER	Outcome
CTC-A	CTC	q,v ($r=8$)	1-layer linear	~ 3.0	~ 0.98	collapse (single token)
CTC-B	CTC	q,k,v,o ($r=16$)	2-layer + trainable LN/CNN	~ 3.0	~ 0.98	collapse (comma/period)
Seq2seq	CE + LS	q,k,v,o ($r=16$)	4-layer Transformer	2.3	0.85	learns Arabic morphemes

5.1.4 Analysis: Why CER Exceeds One

The validation CER and WER in Figure 5.1 exceed 1.0 during the earlier part of training (the Arabic CER ends *below* one, at 0.85). This is not a bug but a consequence of the metric under over-generation: CER is the edit distance normalised by the *reference* length, so when the hypothesis is substantially longer than the reference, the insertions alone can push the ratio above one (Section 2.6). Early in training the decoder has learned the Arabic character prior but not yet a reliable stopping criterion, so it continues emitting plausible characters past the true end of the utterance. As training progresses and the `</s>`-emission behaviour sharpens, hypothesis lengths contract toward the references and the error rates fall. Two inexpensive mitigations (capping generation length as a function of the input length, and reducing the label smoothing that softens the `</s>` decision) further reduce the inflated component of the metric without changing the underlying predictions.

5.1.5 Qualitative Behaviour

The decoder’s outputs become progressively more language-like, which is the clearest evidence that learning is genuine rather than an artefact of the loss. Table 5.2 traces the qualitative progression. Early hypotheses are a single repeated character; by the middle of training they contain frequent, correctly spelled Arabic function words (definite articles and common conjunctions); by step 8000 they consist of well-formed morphemes, even where the content does not yet match the reference. In other words, the model first learns *what Arabic looks like* and only then begins to condition that prior on the (highly ambiguous) visual signal.

Table 5.2: Qualitative progression of decoder output (transliterated/described; native-script examples can be inserted as a figure). Hypotheses move from a single repeated character, to frequent function words, to well-formed morphemes.

Step	Decoder output (character)
1000	A single character repeated (e.g. the definite-article fragment), no word structure.
3000	Short repeated bigrams; the article and a few high-frequency letters appear.
5000	Real frequent words emerge (e.g. “now”, “that”), separated by spaces.
8000	Well-formed Arabic morphemes with valid prefixes; sentence-like spacing.

5.1.6 Comparison with Published Baselines

The MuAViC visual-only Arabic baseline reported in the original corpus paper is on the order of $\sim 77\%$ WER, obtained with *full* encoder–decoder fine-tuning, a pre-trained decoder, and the complete training set [2]. Our configuration (a frozen backbone, only a few million trainable adapter parameters, a from-scratch decoder, and a single-GPU budget) is a strictly harder regime, and the Arabic result (validation CER 0.85, WER ≈ 1.10) remains well above that baseline. The contribution of this thesis is therefore not a new state of the art but a precise characterisation of where parameter-efficient adaptation reaches its ceiling under these combined constraints, and a demonstration that the *choice of training objective* (seq2seq over

CTC) is the decisive factor separating a model that learns nothing from one that learns the language.

5.2 Per-Language Results Across MuAViC

The same recipe analysed for Arabic (the LoRA configuration, decoder architecture, optimisation schedule, and infrastructure of Chapter 4) is applied identically to German and Russian. Across languages only the language-specific normalisation (Section 4.1.2) and the resulting character vocabulary change; the model, objective, and hyperparameters are held fixed. This controlled design isolates the effect of the language and of the amount of per-language data on the behaviour of a single parameter-efficient method.

Table 5.3 consolidates the per-language validation and held-out test error rates, and Figure 5.4 compares them side by side. All three languages were trained for the full 30,000 steps with the identical recipe. On the test set German attains the lowest character error rate (0.79), followed by Russian (0.85) and Arabic (0.87); test word error rates run 1.12 (ar), 1.31 (de) and 1.42 (ru). Test tracks validation closely (within a couple of points, and for German and Russian even slightly below it), which indicates the models did not simply overfit the validation set. All three remain in the high-error regime characteristic of visual-only, low-resource VSR, yet the consistently sub-1.0 test CER confirms that the same recipe transfers across three distinct scripts (Arabic abjad, Latin, Cyrillic) with no per-language tuning.

The gap between CER and WER is widest for Russian and German and smallest for Arabic: characters are partially recovered while whole words rarely are, and longer words and richer morphology amplify each character error at the word level. A firm attribution of the cross-language ordering to data quantity would require controlling for per-language training-set size, which we flag as a confound rather than a conclusion.

Figures 5.2 and 5.3 show the German and Russian training dynamics; both mirror the Arabic trajectory of Figure 5.1: a sharp early loss drop, CER crossing below 1.0, and a WER plateau above it.

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Table 5.3: Per-language results of the proposed method across the three MuAViC languages used here (validation and held-out test CER/WER).

Language	Val CER	Val WER	Test CER	Test WER
Arabic (ar)	0.85	1.10	0.87	1.12
German (de)	0.80	1.37	0.79	1.31
Russian (ru)	0.93	1.62	0.85	1.42

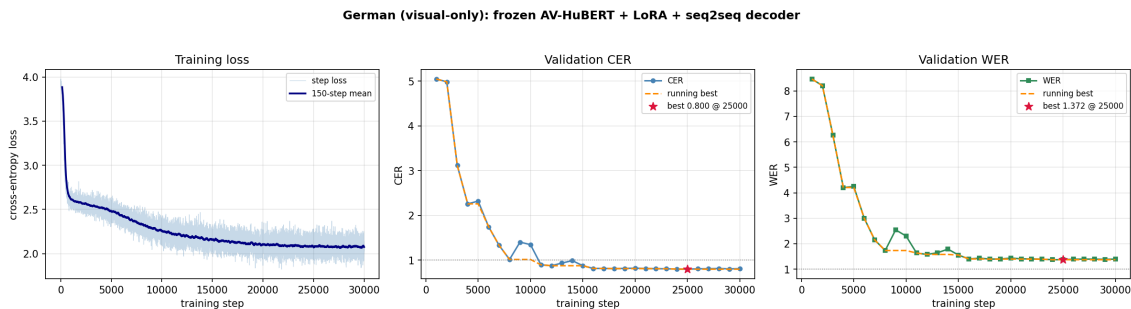


Figure 5.2: German (visual-only): training loss, validation CER, and validation WER. Best validation CER 0.80 at step 25,000.

5.3 Parameter-Efficient vs. Full Fine-Tuning

The preceding sections fix the adaptation method and vary the language; this section fixes the language and varies the *adaptation strategy*, addressing RQ2 directly: how much accuracy does parameter-efficient adaptation concede relative to fully fine-tuning the same backbone? We run a full-fine-tuning baseline identical to our method in every respect (the same AV-HuBERT backbone, the same from-scratch four-layer decoder, and the same data, normalisation, effective batch size, 30,000-step schedule, and greedy decoding), differing only in that the backbone is unfrozen and trained rather than frozen and adapted with LoRA. To keep the comparison fair, the two learning rates are set independently: the pre-trained encoder is fine-tuned at the conventional, gentler rate of 10^{-5} , while the from-scratch decoder retains the 5×10^{-5} used in the LoRA runs, so the component that actually generates the text is neither advantaged nor handicapped by the choice of strategy. What remains is the comparison of interest. The LoRA configuration trains roughly 33M parameters in total (the 3.1M low-rank adapters, the ~ 12 M unfrozen layer-norm and visual/audio front-end parameters, and the 17.6M from-scratch decoder),

5. EXPERIMENTS AND RESULTS

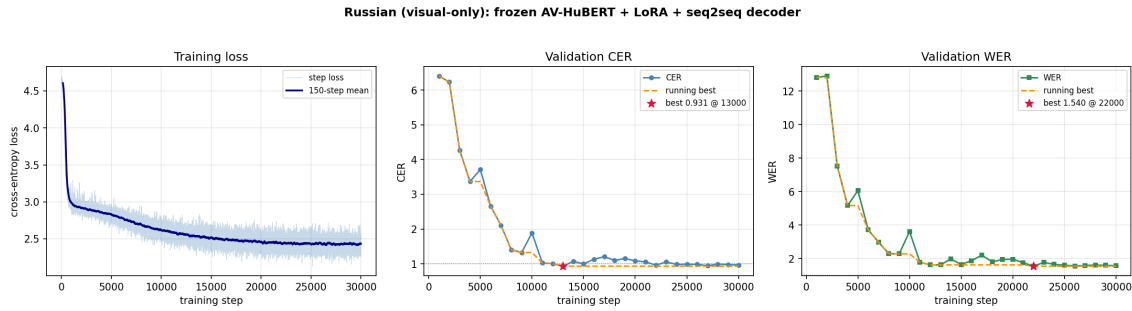


Figure 5.3: Russian (visual-only): training loss, validation CER, and validation WER. Best validation CER 0.93 at step 13,000.

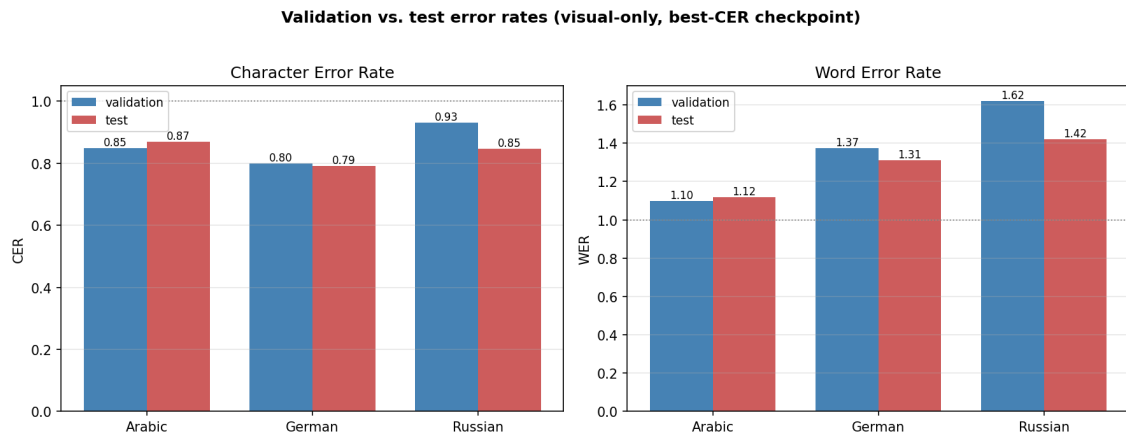


Figure 5.4: Validation vs. held-out test error rates (CER, left; WER, right) for the three languages, at each language’s best-CER checkpoint. Test closely tracks validation, indicating no validation overfitting.

whereas full fine-tuning trains all $\sim 326\text{M}$ backbone parameters plus the same decoder ($\sim 344\text{M}$). The adapters proper are only 3.1M , about 1% of the backbone, so the two approaches differ by roughly an order of magnitude ($\sim 10\times$) in trainable parameters.

Table 5.4 and Figure 5.5 report the result. Full fine-tuning is, at most, only *marginally* better, and the result is closer than expected. On the held-out test set it lowers character error by 0.066 on Arabic ($0.87 \rightarrow 0.80$) and 0.063 on German ($0.79 \rightarrow 0.73$), an 8–9% relative gap, but on Russian the two are level, with LoRA in fact marginally *ahead* on character error (0.85 vs 0.86). Full fine-tuning’s consistent advantage is instead at the word level ($1.12 \rightarrow 1.07$ on Arabic, $1.31 \rightarrow 1.13$ on German, $1.42 \rightarrow 1.33$ on Russian), where the extra capacity helps most; the validation split shows the same pattern. All of this is in exchange for roughly a $10\times$ increase

5. EXPERIMENTS AND RESULTS

in trainable parameters. In short, LoRA recovers essentially all of full fine-tuning’s character-level accuracy (matching or beating it on Russian and coming within $\sim 8\%$ on Arabic and German) while training about a tenth as many parameters.

This is the trade-off at the heart of RQ2. Parameter-efficient adaptation does not *fully* close the gap to full fine-tuning (the larger trainable budget does help), but the residual gap is small in absolute terms and modest in relative terms, and it is purchased at a parameter and optimiser-memory cost roughly an order of magnitude larger. Under the low-resource, single-GPU constraints of this study, LoRA is therefore the more sensible operating point, and this controlled experiment quantifies precisely how much accuracy that choice costs. Two caveats apply. First, this compares *adaptation strategies under a fixed budget*, not best-achievable systems: a longer schedule, stronger regularisation, or per-layer learning-rate tuning could shift the full-fine-tuning figure, albeit at further compute cost. Second, this controlled baseline is distinct from the published MuAViC system discussed in Section 5.1, which uses a pre-trained decoder and the full training set and remains a separate, stronger reference point.

Table 5.4: RQ2: parameter-efficient adaptation (LoRA) vs. full fine-tuning of the same backbone, under an identical recipe and budget. Best-CER validation and held-out test error rates. Full fine-tuning trains all $\sim 326\text{M}$ backbone parameters plus the shared decoder ($\sim 344\text{M}$); the LoRA configuration trains $\sim 33\text{M}$ in total (3.1M adapters plus unfrozen front-ends and the same decoder).

Language	Adaptation	Trainable	Val CER	Val WER	Test CER	Test WER
Arabic (ar)	LoRA ($r=16$)	33M	0.85	1.10	0.87	1.12
	Full fine-tuning	344M	0.77	1.04	0.80	1.07
German (de)	LoRA ($r=16$)	33M	0.80	1.37	0.79	1.31
	Full fine-tuning	344M	0.70	1.11	0.73	1.13
Russian (ru)	LoRA ($r=16$)	33M	0.93	1.62	0.85	1.42
	Full fine-tuning	344M	0.93	1.45	0.86	1.33

5. EXPERIMENTS AND RESULTS

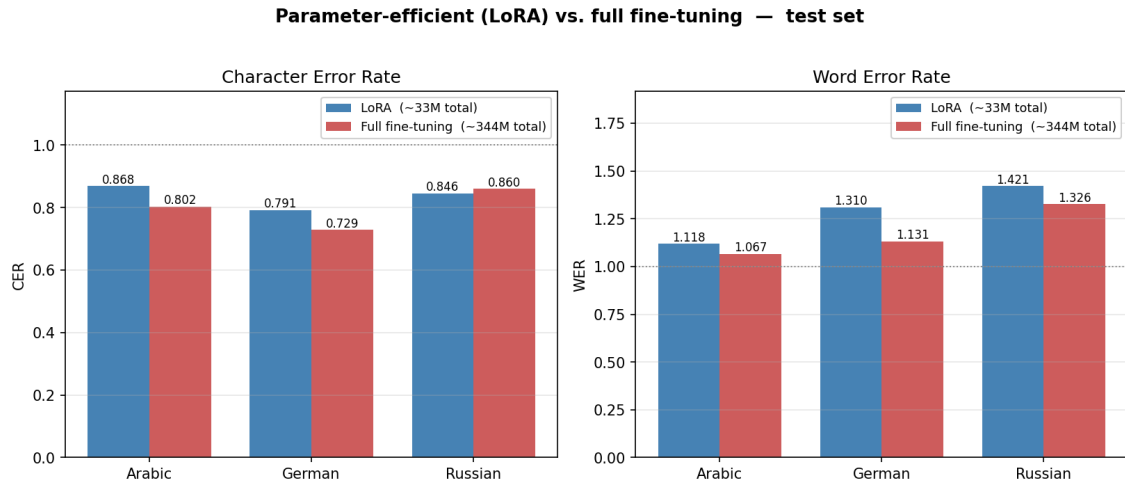


Figure 5.5: RQ2: parameter-efficient adaptation (LoRA, $\sim 33\text{M}$ trainable) vs. full fine-tuning ($\sim 344\text{M}$) of the same backbone, on the held-out test set. Full fine-tuning is marginally better on Arabic and German (about an 8% relative reduction in CER) and level on Russian, while LoRA reaches that accuracy at roughly a tenth of the trainable-parameter cost.

5.4 Data Quality: Filtering Non-Face Training Clips

The results so far take the MuAViC clips as given. Inspecting the Arabic training set, however, shows that a sizeable fraction do not depict the speaker’s mouth at all: because MuAViC is built from TED/TEDx recordings, the cropped tracks include title cards, wide and stage shots, and cuts to the audience, each carrying a transcript but no readable lip movement. Training on these asks the model to map non-speech imagery to characters, which is *active* label noise rather than mere data scarcity.

We quantify it with a per-clip *face fraction*: each clip is sampled at several frames, a frame counts as a mouth when a YCrCb skin/lip test exceeds a small threshold, and the face fraction is the proportion of such frames. The distribution is strongly bimodal (most clips score near 1.0, while a clear $\sim 12\text{--}16\%$ score near 0), so a cut at 0.6 falls in an empty valley rather than through a dense region. Discarding clips below it (Figure 5.6), together with a few character-rate outliers, removes 16% of the Arabic training set, leaving 8,524 of $\sim 10,200$ clips. Only the *training* split is filtered; validation and test are left intact, so the comparison is like-for-like.

5. EXPERIMENTS AND RESULTS



(a) Clips *removed* by the filter (lowest face fraction): TED title cards, wide and stage shots, and cuts to the audience, none showing a readable mouth.



(b) A random sample of *retained* clips: almost all are genuine mouth ROIs.

Figure 5.6: Qualitative examples from the Arabic training set; each tile is one sampled frame (the small green number is the clip index). The face-fraction filter discards the non-face clips in (a) while keeping the lip-ROI clips that dominate the corpus, sampled in (b).

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Retraining the identical LoRA recipe on the filtered set lowers error on both splits *despite* the smaller training set (Table 5.5, Figure 5.7). Validation CER falls from 0.85 to 0.80 and held-out test CER from 0.87 to 0.83 (a 5–6% relative reduction), with word error also lower. Put plainly, *cleaner data beat more data*: roughly a sixth of the training examples were actively harmful, and a cheap, fully automatic audit recovers a meaningful slice of error that additional adaptation capacity had not closed. The audit is language-agnostic and applies unchanged to German and Russian, and label-level cleaning (cross-checking each transcript against its audio track) is a natural next step.

Table 5.5: Effect of removing non-face training clips on Arabic, under the identical LoRA recipe (validation and held-out test at the best-CER checkpoint). Filtering discards 16% of the training clips yet lowers error on both splits.

Training set	Train clips	Val CER	Val WER	Test CER	Test WER
Baseline (all clips)	~10,200	0.85	1.10	0.87	1.12
Filtered (non-face removed)	8,524	0.80	1.07	0.83	1.10

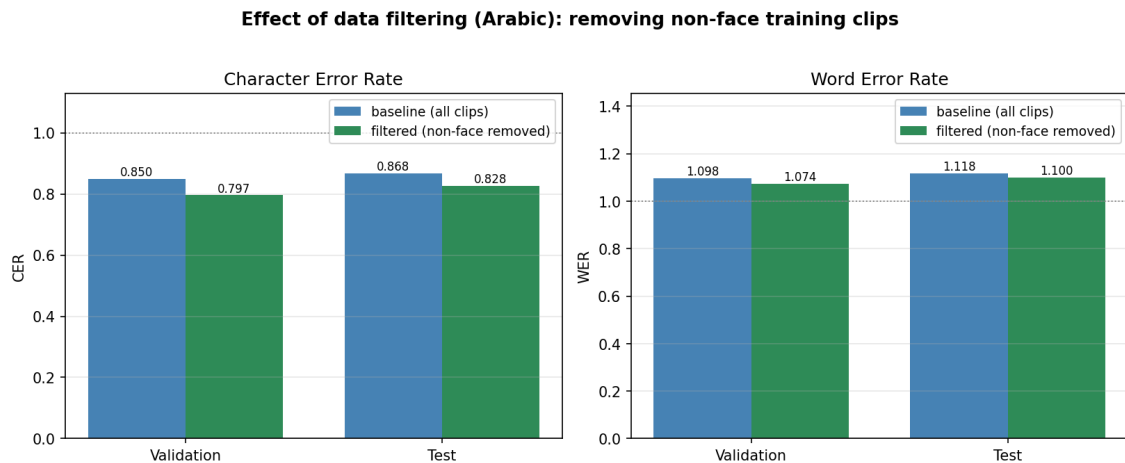


Figure 5.7: Arabic CER and WER before and after removing non-face training clips. Filtering 16% of the training set (TED title cards, wide shots, audience cuts) lowers both validation and test error despite training on fewer clips.

5.5 Decoding, Capacity, and the Performance Ceiling

Both error rates sit above 1.0 for much of training, and word error stays above one even at convergence (Table 5.5). This is the signature of *over-generation*: the decoder emits more words than the reference, so insertions alone push the rate past one, and greedy decoding cannot undo it. Replacing greedy search with beam search (width 5) and a GNMT length penalty ($\alpha = 0.6$, mildly favouring shorter hypotheses) needs no retraining and targets the inflation directly: on the filtered Arabic model it lowers test CER from 0.83 to 0.78 and WER from 1.10 to 1.07. Cumulatively, filtering and beam search move the Arabic test result from the original 0.87/1.12 to 0.78/1.07 (about a 10% relative reduction in character error) at no additional training cost.

Neither change brings word error below one, which raises the question of whether the remaining error is a matter of model capacity. We tested this directly: doubling the adapter rank ($r=16 \rightarrow 32$, with α scaled to preserve the update magnitude) and sharpening the objective (label smoothing $0.1 \rightarrow 0.05$), retrained on the same filtered data. The larger model fit the training set *better* (final loss $2.32 \rightarrow 2.18$) yet generalised *worse*: validation CER rose from 0.80 to 0.84, and the test result was unchanged (0.79/1.06). That is overfitting, not improvement: additional adaptation capacity is not the binding constraint.

Inspecting the decoder’s output explains why. Across the test set the hypotheses are fluent but *generic* and nearly identical from clip to clip: each tends to open with the same high-frequency word and is assembled from common function words (articles, conjunctions, “but”, “which”, “cannot”), with little dependence on the particular video. The model has acquired a strong prior over *what the language looks like*, but the visual-only signal in this low-resource setting is too weak and too ambiguous to steer that prior toward the actual utterance, since many phonemes are visually identical (homophenes). The same effect underlies the RQ2 result of Section 5.3: fully fine-tuning the entire ~ 326 M backbone reaches essentially the same word error (≈ 1.07) as the three-million-parameter adapter. When a small adapter,

a doubled adapter, and the full model all converge to the same word error (Table 5.6), the binding constraint is the *task* (visual-only lip-reading of a low-resource language), not the adaptation method, its capacity, or the decoder. This bounds what the present recipe can achieve and motivates the directions in Chapter 6: initialising the decoder from a pre-trained language model and adding in-domain data, rather than enlarging the adapter.

Table 5.6: Arabic held-out test error across the improvement steps. Gains come from data quality and decoding and then plateau; full fine-tuning, with two orders of magnitude more trainable parameters, reaches the same word error (greedy decoding).

Configuration	Test CER	Test WER
Baseline (all clips, greedy)	0.87	1.12
+ data filtering (greedy)	0.83	1.10
+ beam search ($r=16$)	0.78	1.07
+ rank 32, label smoothing 0.05	0.79	1.06
Full fine-tuning (reference, greedy)	0.80	1.07

5.5.1 Cross-Language Generalisation

The improvements above are not specific to Arabic. Applying the same pipeline (data filtering, the rank-32 adapter, and beam decoding with a length penalty) to German and Russian lowers held-out test error for all three languages (Table 5.7, Figure 5.8). The gains are largest in *word* error, where over-generation had inflated the baseline most: German falls from 1.31 to 1.07 and Russian from 1.42 to 1.17 (both $\approx 18\%$ relative), driven chiefly by beam search. Character error also improves everywhere, substantially for Arabic and German ($0.87 \rightarrow 0.79$ and $0.79 \rightarrow 0.73$) and marginally for Russian ($0.85 \rightarrow 0.84$). The higher-rank adapter behaves inconsistently across languages: it helps German but, as on Arabic (Section 5.5), tends to overfit the smaller partitions. The reliable, language-agnostic drivers are therefore cleaner data and better decoding rather than adapter capacity. All three none the less remain above the 1.0 word-error mark: the pipeline narrows the gap to the ceiling without breaking through it.

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Table 5.7: Effect of the improvement pipeline (data filtering, rank-32 adaptation, beam decoding) on held-out test error, for all three languages. Error falls in every case, most strongly in word error.

Language	Baseline		Improved	
	CER	WER	CER	WER
Arabic (ar)	0.87	1.12	0.79	1.06
German (de)	0.79	1.31	0.73	1.07
Russian (ru)	0.85	1.42	0.84	1.17

Cross-language effect of the improvement pipeline (held-out test)

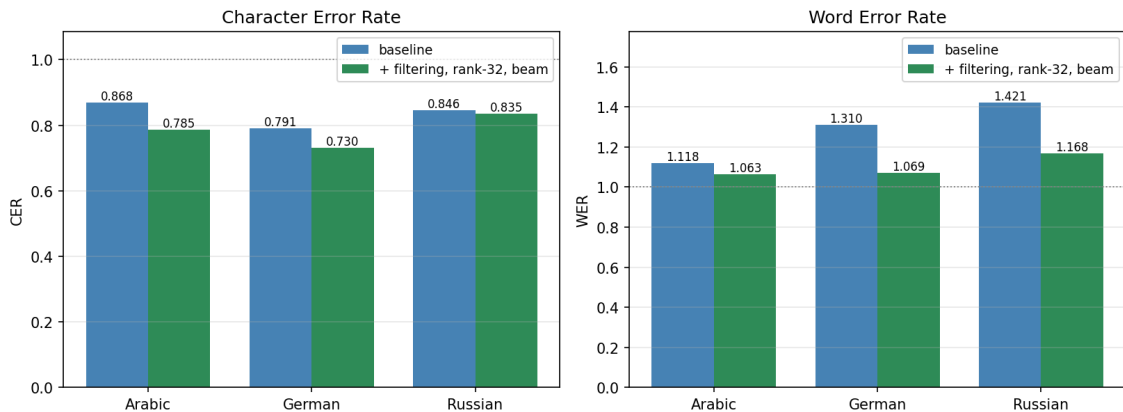


Figure 5.8: Baseline vs. the improvement pipeline (data filtering, rank-32 adaptation, beam decoding) on the held-out test set. Test error falls for all three languages, with the largest reductions in word error.

5.6 Value of the Self-Supervised Backbone

Every result so far builds on the frozen, pre-trained AV-HuBERT backbone. How much does that pretraining actually contribute? To answer this we train, for each language, an *otherwise identical* model (the same filtered training data, same character decoder, same label-smoothed cross-entropy, same greedy/beam evaluation) but replace the pre-trained backbone with a visual encoder initialised *from scratch*: a standard lip-reading frontend (3D-convolutional stem + ResNet-18 per frame + a six-layer Transformer, ~ 45 M parameters), trained end-to-end. Only the origin of the encoder changes, so the comparison isolates the value of self-supervised pretraining (RQ1).

Table 5.8 and Figure 5.9 report the outcome, and the same two-sided pattern holds for all three languages. The pre-trained backbone gives a large *head start*

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before decoding is tuned: on every language it lowers validation and greedy test CER substantially, because the from-scratch model over-generates badly (its validation CER reaches 0.95 on German and 1.09 on Russian, against 0.77 and 0.88 for the pre-trained model). Beam search with a length penalty, which corrects exactly that over-generation, *nearly erases the gap*: on the held-out test set the from-scratch encoder comes within 0.01–0.02 CER of the pre-trained backbone (Arabic 0.80 vs 0.78, German 0.76 vs 0.75, Russian 0.78 vs 0.78) and in fact *beats* it on word error for all three (1.05 vs 1.07, 1.09 vs 1.12, and 0.99 vs 1.02). Pretraining therefore helps, and helps most *before* decoding is tuned, but it does not raise the ceiling: a 45M encoder trained from nothing reaches the same plateau as the 326M pre-trained backbone adapted with a 3M adapter, because both ultimately fall back on the language prior once the visual-only signal runs out. The self-supervised backbone buys faster, cleaner convergence, not a higher ceiling.

Table 5.8: Value of self-supervised pretraining across the three languages: an otherwise identical model with a from-scratch visual encoder instead of the frozen pre-trained backbone, both trained on the filtered data. The backbone helps most on validation and greedy decoding; beam search nearly closes the CER gap, and the from-scratch model even wins on word error. Best value per language and metric in bold.

Language	Encoder	Val CER	Test CER (greedy)	Test CER (beam)	Test WER (beam)
Arabic	Pre-trained AV-HuBERT + LoRA	0.80	0.83	0.78	1.07
	From-scratch encoder	0.89	0.93	0.80	1.05
German	Pre-trained AV-HuBERT + LoRA	0.77	0.78	0.75	1.12
	From-scratch encoder	0.95	0.92	0.76	1.09
Russian	Pre-trained AV-HuBERT + LoRA	0.88	0.79	0.78	1.02
	From-scratch encoder	1.09	0.89	0.78	0.99

5.7 Limitations

Several limitations frame the results. (i) Absolute error remains high: visual-only, low-resource VSR is far from audio-visual or fully fine-tuned systems, so the contribution is the parameter-efficient *method* and its cross-lingual characterisation, not state-of-the-art recognition. (ii) The training loss had largely plateaued by 30,000 steps, so only marginal further gains are expected from longer training. (iii) The in-depth objective comparison (CTC vs. seq2seq) is presented on Arabic as a

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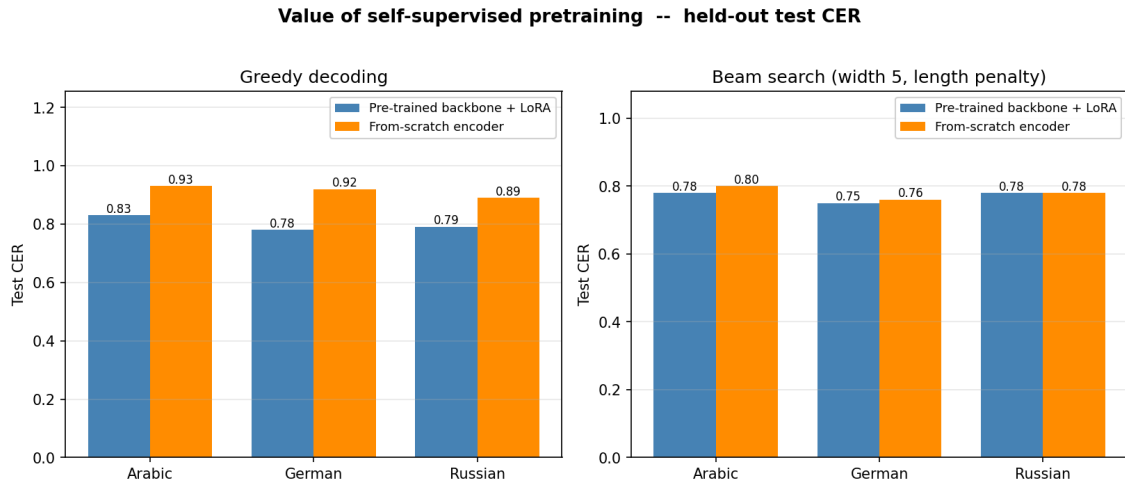


Figure 5.9: Value of self-supervised pretraining across the three languages: the pre-trained backbone versus a from-scratch encoder, otherwise identical, on held-out test CER. Under greedy decoding (left) the backbone leads on every language; beam search (right) almost closes the gap, so the backbone’s advantage is in convergence rather than the achievable ceiling.

representative language; the per-language error rates are consolidated in Table 5.3.

(iv) The residual error is bounded by the weak visual signal rather than by decoding or adapter capacity (Section 5.5): beam search and data filtering both lowered the error rates, but word error remains around one because the decoder leans on a language prior that the visual stream cannot reliably steer.

5.8 Summary of Results

Table 5.9 consolidates the held-out test error of every configuration across the three languages. Two patterns stand out. First, the improvement pipeline (data filtering, then beam search) lowers error for all three languages over the unfiltered greedy baseline. Second, the best result in each language is reached by a *different* method (LoRA with filtering and beam search, full fine-tuning, or the from-scratch encoder), and the spread between them is small: once the visual signal is exhausted, the adaptation strategy matters little. This is the clearest single view of the performance ceiling that runs through the chapter.

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Table 5.9: Master summary: held-out test CER/WER for every configuration across the three MuAViC languages, with the mean over the three languages. The lowest CER and WER in each column are in **bold**. Decoding is greedy except for the two beam-search rows.

Configuration	Trainable	Arabic		German		Russian		Average	
		CER	WER	CER	WER	CER	WER	CER	WER
LoRA, baseline (greedy)	33M	0.87	1.12	0.79	1.31	0.85	1.42	0.84	1.28
LoRA, filtered (greedy)	33M	0.83	1.10	0.78	1.37	0.79	1.18	0.80	1.22
LoRA, filtered + beam	33M	0.78	1.07	0.75	1.12	0.78	1.02	0.77	1.07
Full fine-tuning (greedy)	344M	0.80	1.07	0.73	1.13	0.86	1.33	0.80	1.18
From-scratch encoder + beam	45M	0.80	1.05	0.76	1.09	0.78	0.99	0.78	1.04

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Summary

This thesis examined whether a frozen, self-supervised audio-visual foundation model can be adapted for visual-only speech recognition using only parameter-efficient techniques, and whether a single such recipe generalises *across languages*. A systematic and critical review of 42 papers established that the paradigm (a frozen backbone plus lightweight adapters) is well supported in the literature across expressiveness, efficiency, decoder minimalism, robustness, and low-resource transfer, but also that this support was gathered under favourable conditions (English, audio-visual input, abundant data) that are never removed simultaneously; the considerations distilled from that critique directly shaped the design. We then defined a uniform adaptation recipe and a multilingual evaluation protocol over three languages of MuAViC (Arabic, German, Russian), under deliberately adverse conditions: visual-only input, a frozen AV-HuBERT backbone adapted with LoRA, a lightweight decoder, and a single-GPU budget.

Using Arabic (a demanding low-resource partition with a non-Latin abjad script) as a detailed case study, the empirical work yielded a clear, two-part finding. A CTC head over the frozen features *collapses* to a degenerate single-token solution

that no amount of learning-rate, rank, head-depth, or layer-norm tuning could rescue, because the objective’s conditional-independence assumption is pathological for ambiguous visemes in a low-data regime. Replacing CTC with a small sequence-to-sequence decoder trained with label-smoothed cross-entropy *restores learning*: the loss falls from 3.9 to 2.3, the validation CER drops by almost an order of magnitude (to 0.85 on Arabic), and the decoder progressively emits valid Arabic morphemes. The same recipe is applied unchanged to German and Russian. A series of controlled experiments then probes the residual error: full fine-tuning proves only marginally better than LoRA, data filtering and beam search reduce error by a further $\approx 10\%$, yet extra adapter capacity, full fine-tuning, and even an encoder trained *from scratch* all converge to the same ≈ 0.78 CER / 1.05 WER. The binding constraint is therefore not adaptation capacity but the weak visual-only signal, which drives the decoder onto a language prior it cannot reliably steer: the training objective and the task, not the method, set the ceiling.

6.2 Answers to the Research Questions

- **RQ1 (expressiveness).** The frozen backbone does retain decodable information: once paired with an appropriate decoder, it supports learning of Arabic character sequences from video alone, and a comparison against an encoder trained *from scratch* (Section 5.6) confirms the self-supervised pretraining contributes real value (faster, cleaner convergence), even if it does not raise the ultimate ceiling. However, the visual-only signal is markedly less expressive than the audio-visual case the literature usually reports.
- **RQ2 (PEFT vs. full fine-tuning).** A controlled comparison (Table 5.4) shows that fully fine-tuning the ~ 326 M backbone is at most *marginally* better than LoRA (about an 8% relative reduction in test CER on Arabic and German, level on Russian), despite training roughly $10\times$ more parameters. LoRA closes that gap almost entirely, and it recovers essentially all of full fine-tuning’s character-level accuracy (matching it on Russian) while training about a tenth as many parameters (~ 33 M vs ~ 344 M), which makes parameter-efficient adap-

tation the sensible operating point under our low-resource, single-GPU budget.

- **RQ3 (minimalist decoders).** A lightweight decoder is sufficient *provided the training objective is appropriate*; the decisive design choice was seq2seq over CTC, not decoder size.
- **RQ4 (robustness).** Not directly evaluated in our experiments; addressed only through the literature review.
- **RQ5 (low-resource and cross-lingual).** The method learns even in the low-resource Arabic setting; data filtering and beam search improve every language (Section 5.5.1) but do not lift word error below one, and absolute error rates remain uncompetitive; they are bounded, as the ceiling analysis shows, by the visual-only signal rather than by data quantity alone. The cross-lingual behaviour is quantified by the per-language comparison (Table 5.3).

6.3 Future Work

Several directions would strengthen and extend the cross-corpus study:

- **Decoder pre-training.** Initialising the decoder from a pre-trained language model, rather than from scratch, is the most likely single change to close the gap with the MuAViC baselines, and is especially promising for the higher-resource languages.
- **Stronger visual representations.** Because the ceiling is set by the visual signal rather than the adaptation method, richer visual front-ends or additional in-domain pretraining data are more promising than further adapter capacity, which was shown to overfit the smaller partitions.
- **Joint multilingual training** with shared adapters, to test whether a single set of LoRA parameters can serve several languages at once and whether cross-lingual transfer benefits the smaller partitions.
- **Label-level data cleaning**, cross-checking each transcript against its audio track to remove the mislabelled clips that the face-based filter cannot detect.

- **Robustness evaluation** (occlusion, noise, pose) to test RQ4 empirically rather than only through the literature.

6.4 Closing Remark

The practical lesson of this work is that parameter-efficient adaptation of a frozen backbone is necessary but not sufficient for visual speech recognition: the training objective and the decoder formulation determine whether the frozen features can be exploited at all. Establishing that boundary on the most challenging language, fixing a recipe that works under realistic compute, and demonstrating that the same recipe transfers unchanged to two further languages is the contribution offered here, together with an honest characterisation of the ceiling that visual-only, low-resource recognition imposes on any adaptation method.

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